

Multiplicity Obstructs a Universal Orthogonal Decomposition: An S_3 Counterexample to Conjecture 1.1

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Status. Candidate counterexample, likely valid, for human review.

1 Source conjecture

Hokamp's Conjecture 1.1 asks whether, whenever a compact group G acts continuously and transitively on a compact Hausdorff space X , there is one collection $\mathcal{G}$ of closed, pairwise L^2 -orthogonal, G -minimal subspaces of $C(X)$ such that every uniformly closed G -invariant subspace of $C(X)$ is the uniform closure of the direct sum of a subcollection of $\mathcal{G}$ [1, Conjecture 1.1, PDF p. 1].

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In [4], Nagel and Rudin establish a decomposition theorem for the closed unitarily invariant spaces of continuous and measurable functions on the unit sphere of $\mathbb{C}^n$. This work can also be found in [2]. Using the work of Nagel and Rudin as an inspiration, we conjecture the following. Note that $C(X)$ denotes the continuous complex functions with domain X .

Conjecture 1.1. Let G be a compact group acting continuously and transitively on a compact Hausdorff space X . Then there exists a collection $\mathcal{G}$ of closed, pairwise orthogonal, G -minimal subspaces of $C(X)$ induced by the group action such that any closed G -invariant subspace of $C(X)$ is the closure of the direct sum of some subcollection of $\mathcal{G}$.

Definitions for G -minimal and G -invariant can be found in Section 2.

The existence of a collection $\mathcal{G}$ with the desired qualities is a result of the Peter-Weyl Theorem from [3]. This is shown in Section 4 (Theorem 4.1). Recently in [4], the author proved Conjecture 1.1 subject to a condition on the collection $\mathcal{G}$. An attempt to drop this condition led to the study of the titular class of homeomorphisms on X and the variety of subsequent results established in this paper.

Section 3 is devoted to defining the class of homeomorphisms, establishing their properties, and proving some immediate results, the most important of which is the following:

Here $C(X)$ consists of complex-valued functions, and orthogonality is with respect to the invariant probability measure on X .

2 Claimed contribution

Theorem 1 (Exact obstruction). *Let a compact group G act continuously and transitively on a compact Hausdorff space X , and let U be the resulting unitary representation on $L^2(X)$. The conclusion of Conjecture 1.1 holds for (G, X) if and only if U is multiplicity-free.*

Equivalently, after writing $X \simeq G/K$ for a stabilizer K , the conclusion holds if and only if

$$\dim V_\pi^K \leq 1$$

for every irreducible unitary representation V_π of G .

Corollary 1 (Finite counterexample). *Conjecture 1.1 is false. Take $G = S_3$, let $X = S_3$ with the discrete topology, and let G act on X by left translation. This is a continuous transitive action of a compact group on a compact Hausdorff space, but no collection $\mathcal{G}$ having the asserted universal property exists.*

3 Proof intuition

The conjecture asks a single orthogonal list of irreducible subspaces to generate *every* invariant subspace merely by selecting entries from that list. This is possible when every irreducible type occurs once: the isotypic pieces are then themselves irreducible, and Peter–Weyl spectral projections recover every invariant subspace from a subset of them.

Multiplicity makes the requested list impossible. In two equivalent copies $V \oplus V$, both the first coordinate $V \oplus 0$ and the diagonal $\{(v, v) : v \in V\}$ are minimal invariant subspaces, but they are not orthogonal. The universal property would force *every* minimal invariant subspace to be an entry of the same pairwise-orthogonal list. The left regular representation of S_3 contains two copies of its standard two-dimensional representation, so it supplies a six-point counterexample.

4 Proof of the characterization

We use the following immediate consequence of the word “every” in the source conjecture.

Lemma 1 (Minimal spaces must be listed). *Suppose a collection $\mathcal{G}$ has the universal property in Conjecture 1.1. Then every nonzero closed G -minimal subspace $M \subset C(X)$ is itself a member of $\mathcal{G}$.*

Proof. By the universal property, M is the closure of the direct sum of a subcollection $\mathcal{S} \subset \mathcal{G}$. The subcollection is nonempty, and each $H \in \mathcal{S}$ is contained in that direct sum and hence in M . Since H is nonzero and G -invariant while M is minimal, $H = M$. Thus $M \in \mathcal{G}$. $\square$

Proof of Theorem 1. Let μ be the invariant probability measure. Peter–Weyl theory gives the isotypic decomposition

$$L^2(X, \mu) = \widehat{\bigoplus_{\pi \in \widehat{G}} V_\pi \otimes M_\pi},$$

where M_π is the multiplicity space [2, 3]. Because $X \simeq G/K$, Frobenius reciprocity identifies M_π with $(V_\pi^*)^K$, so $\dim M_\pi = \dim V_\pi^K < \infty$.

The finite-dimensional isotypic space $H_\pi = V_\pi \otimes M_\pi$ consists of continuous functions. One direct way to see this is to use the character projection

$$P_\pi f = d_\pi \int_G \overline{\chi_\pi(g)} U_g f dg.$$

It maps $C(X)$ to $C(X)$ and is the L^2 -orthogonal projection onto H_π . Since $C(X)$ is dense in $L^2(X)$ and H_π is finite dimensional, $P_\pi(C(X)) = H_\pi$.

Assume first that some $\dim M_\pi \geq 2$. Choose orthonormal vectors $e_1, e_2 \in M_\pi$. Inside $C(X)$ consider

$$H_1 = V_\pi \otimes \mathbb{C}e_1, \quad H_\Delta = V_\pi \otimes \mathbb{C}(e_1 + e_2).$$

Both are closed G -minimal subspaces. They are distinct and not orthogonal, since for $0 \neq v \in V_\pi$,

$$\langle v \otimes e_1, v \otimes (e_1 + e_2) \rangle = \|v\|^2 \neq 0.$$

If the conjectured family existed, Lemma 1 would put both H_1 and H_Δ into one pairwise-orthogonal family, a contradiction. Thus the conjecture implies multiplicity-freeness.

Conversely, suppose $\dim M_\pi \leq 1$ for every π . Each nonzero H_π is then G -minimal, and the collection of these H_π is pairwise orthogonal. Let $Y \subset C(X)$ be uniformly closed and G -invariant. The Bochner-integral formula above shows that $P_\pi Y \subset Y$. Irreducibility therefore gives either $P_\pi Y = 0$ or $P_\pi Y = H_\pi$.

Finally, the Peter–Weyl finite-spectrum vectors are uniformly dense in every continuous Banach representation of a compact group. Concretely, a net of central trigonometric-polynomial approximate identities gives operators

$$T_\alpha y = \int_G k_\alpha(g) U_g y \, dg$$

that converge to y in the uniform norm and are finite linear combinations of the projections P_π . Since Y is invariant and closed, this applies inside Y . Hence

$$Y = \overline{\bigoplus_{\{\pi: P_\pi Y = H_\pi\}} H_\pi}^{\|\cdot\|_\infty},$$

which is exactly the asserted universal-subcollection property. The equivalence with $\dim V_\pi^K \leq 1$ was supplied by Frobenius reciprocity. $\square$

5 An explicit S_3 witness

Although Theorem 1 immediately proves Corollary 1, the following construction makes the obstruction fully concrete. Put

$$V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : z_1 + z_2 + z_3 = 0\},$$

with S_3 permuting coordinates. This is the irreducible standard two-dimensional representation. For example, with

$$a = \frac{(1, -1, 0)}{\sqrt{2}}, \quad b = \frac{(1, 1, -2)}{\sqrt{6}},$$

the transpositions (12) and (23) have matrices

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

They have no common invariant line, proving irreducibility.

Equip $C(S_3)$ with the uniform-probability inner product

$$[f, k] = \frac{1}{6} \sum_{g \in S_3} f(g) \overline{k(g)},$$

and write ρ for the standard representation. For nonzero $q \in V$, define

$$(F_q v)(g) = \langle v, \rho(g)q \rangle, \quad M_q = F_q(V) \subset C(S_3),$$

using the convention that $\langle \cdot, \cdot \rangle$ is linear in its first argument. For the left regular action $(U_h f)(g) = f(h^{-1}g)$,

$$U_h F_q v = F_q(\rho(h)v).$$

Schur orthogonality gives, since $\dim V = 2$,

$$[F_q v, F_r w] = \frac{1}{2} \langle v, w \rangle \langle r, q \rangle. \quad (1)$$

In particular, F_q is injective, so M_q is a closed minimal invariant copy of V .

Let $c = a + b$. Formula (1) shows that $M_a \perp M_b$, while

$$F_c v = F_a v + F_b v, \quad [F_a v, F_c v] = \frac{1}{2} \|v\|^2 \neq 0 \quad (v \neq 0).$$

Thus M_a and M_c are nonorthogonal. They are distinct: if $M_c = M_a$, then $F_b v = F_c v - F_a v$ would put the nonzero space M_b inside M_a , contradicting $M_a \perp M_b$. Lemma 1 says that a universal family would have to contain both M_a and M_c , contradicting pairwise orthogonality. This proves Corollary 1 directly.

6 Verification, novelty, and scope

The exact symbolic checker in `code/verify_s3_counterexample.py` enumerates all six permutations, constructs the coefficient spaces M_a, M_b, M_c , verifies their ranks and invariance, checks irreducibility via the commutant of the generating transpositions, and confirms $M_a \perp M_b$ but $M_a \not\perp M_c$ with $M_a \neq M_c$. All checks pass. The computation only verifies the displayed finite example; the proof above is exact and does not depend on numerical evidence.

A bounded novelty search on 11 August 2026 checked the run’s lightweight indexes; the exact arXiv id and title; the author’s name together with the conjecture; the phrases “pairwise orthogonal G -minimal” and “multiplicity-free”; arXiv:2308.09799; and the earlier conditional paper arXiv:2110.12060. The source remains arXiv v1, submitted 18 August 2023. No later paper explicitly proving or disproving Conjecture 1.1 was found. The mechanism itself is a direct application of standard compact-group representation theory, so novelty confidence is moderate rather than high.

The packet settles the conjecture as stated and also characterizes exactly which compact homogeneous actions satisfy it. It does not address modified statements that restrict the class of invariant subspaces or permit the orthogonal family to depend on the subspace being represented.

Human-review recommendation. Check first the short minimal-space Lemma 1 and the three coefficient spaces in the S_3 construction. The general characterization can then be audited through the standard isotypic projection and finite-spectrum density steps.

References

- [1] S. A. Hokamp, *A Class of Homeomorphisms on Homogeneous Spaces of a Group Action*, arXiv:2308.09799, 2023.
- [2] F. Peter and H. Weyl, *Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe*, Math. Ann. 97 (1927), 737–755.
- [3] G. B. Folland, *A Course in Abstract Harmonic Analysis*, CRC Press, 1995.